

ECOSYSTEM API

Connection Guide for Service Providers

This guide is the primary technical reference for service providers joining the Equus Ecosystem using the API Method of connecting. It provides an overview of the platform and its components, the integration and standard data flows between web services and AssignmentPro, an introduction to the Ecosystem-related data sets, and an overview of the process for onboarding and connecting service providers who are new to the Ecosystem.

This document is meant to be accompanied by supporting documents, which include a guide to supported web service authentication methods and one or more service provider service-specific supplementary technical documents.

CONTENTS

1	The Equus Ecosystem Platform and Its Components	1
1.1	Equus AssignmentPro Instances.....	2
1.2	The Equus Hub.....	3
1.2.1	API Method of Connection.....	3
1.2.2	Other Methods of Connection.....	3
1.2.3	Traffic Flow and Data Exchange	4
1.3	Service Provider Systems.....	5
1.3.1	Commonly Requested Service Types	5
1.3.2	Communication Endpoints.....	7
1.3.3	Connections to AssignmentPro	7
2	Ecosystem Integration Between Service Provider and AssignmentPro.....	8
2.1	AssignmentPro to Service Provider	9
2.1.1	Standard Data Flows	9
2.1.2	Request and Response Communication	11
2.2	Service Provider to AssignmentPro	14
2.2.1	Standard Data Flows	14
2.2.2	Request and Response Communication	15
2.2.3	Error Responses and Troubleshooting.....	17
3	Ecosystem-Related Data Sets	23
3.1	Recommended Service	23
3.2	Service Provider-Agreed.....	23
3.3	Client/Service Provider-Agreed	23
3.4	Permission Sets.....	24
4	Service Provider Onboarding and Connection Process.....	25
4.1	New Ecosystem Service Provider Onboarding.....	25
4.1.1	New Service Provider Kickoff Meeting	25
4.1.2	Data Mapping Collaboration	26
4.1.3	Development.....	26
4.1.4	Initial Service Provider Connection and Testing.....	26
4.2	Service Provider to Client AssignmentPro Instance Connection and Testing.....	27
4.2.1	Pre-Kickoff Meeting	27
4.2.2	Client/Service Provider Kickoff Meeting	27
4.2.3	Data Mapping Collaboration	27
4.2.4	Initial Client Instance Connection and Testing	27

EXHIBITS

Exhibit 1: The Equus Hub manages secure communication	3
Exhibit 2: Standard Ecosystem data flows.....	8
Exhibit 3: Example service order within AssignmentPro.....	9
Exhibit 4: Example request header	12
Exhibit 5: Example request payload.....	13
Exhibit 6: Example required incoming headers.....	15
Exhibit 7: Example JSON script to set the vendor status field to "Accepted"	16
Exhibit 8: Example of a successfully processed request	17
Exhibit 9: Standard HTTP connection error codes	18
Exhibit 10: Example "Access Denied" or "Non-Existing Record" error message	18
Exhibit 11: Example CRUD error message.....	19
Exhibit 12: Example EPIC Syntax/JSON Mapping error message	20
Exhibit 13: Example JSON Syntax error message	21
Exhibit 14: Example error message for a missing required field in a JSON command script	21
Exhibit 15: Example error message for an invalid lookup value	22

1 The Equus Ecosystem Platform and Its Components

The **Equus Ecosystem** (Ecosystem) is a data communication platform designed to connect the clients of Equus' award-winning portfolio of corporate workforce mobility products, including AssignmentPro, with providers of the mobility-related services on which those organizations and their mobile employees rely.

Providers of mobility-related services who connect to the Ecosystem can benefit from streamlined communications, more automated workflows, less information and document requests, improved data integrity, and increased client satisfaction. These benefits improve the results of key performance indicators and SLA metrics as well as add client value.



 For more information, see the [Equus Ecosystem Brochure](#).⁷

The Ecosystem platform supports the assignment lifecycle by allowing data to be shared in real-time between AssignmentPro and service providers' business technology tools. The platform consists of three key components:



These essential components of the Ecosystem platform are detailed in the following sections.

1.1 Equus AssignmentPro Instances

Equus AssignmentPro (AssignmentPro) is a global workforce management tool that covers the complete expatriate lifecycle, from pre-authorization and documentation to relocation, compensation, taxes, and repatriation.

AssignmentPro helps organizations handle the complexities involved with ordering and tracking the services required by their mobile employees. AssignmentPro is used by corporate mobility programs and relocation and assignment management companies that provide those services to their clients. AssignmentPro is typically offered as Software as a Service (SaaS) but may also be installed on-premise.

 For more information, see the [Equus AssignmentPro Brochure](#).⁷

The visibility of the information (how much and what is seen) within AssignmentPro is very customizable and depends on a variety of factors—the primary factor being the user type:

- **Mobility Team.** The mobility team has the most visibility into the information within AssignmentPro. They have an overarching view of everything needed to initiate, organize, and manage employee assignments and all associated tasks, activities, and communication.
- **Human Resources Business Partners (HRBPs) and Other Managers.** Many clients using AssignmentPro are given



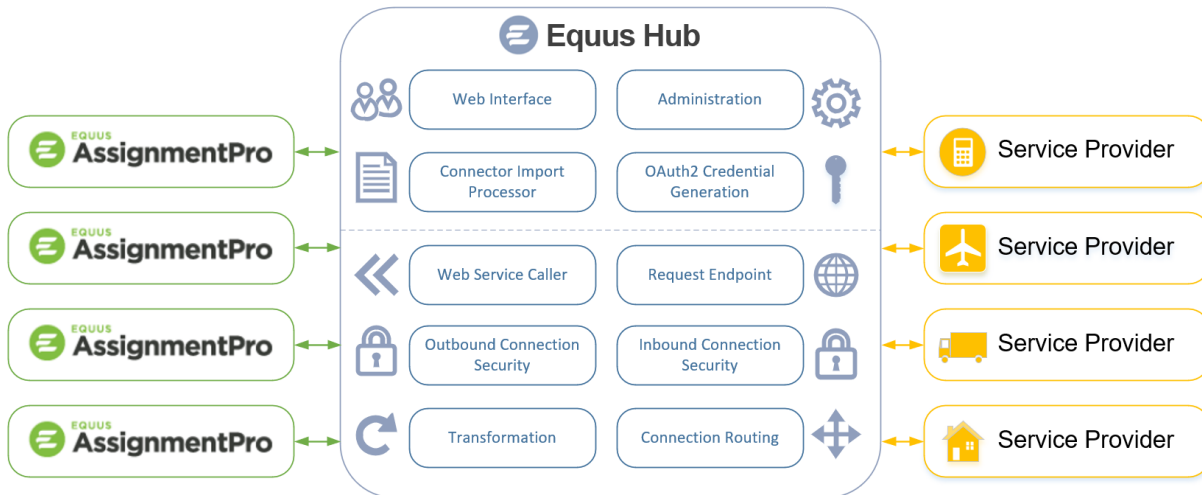
access to information on a subset of employees they might manage. The amount of visibility and interaction these users have depends on instance configuration and such factors as the stage of the assignment. These users may need to order services for the assignee but may not have visibility into the downstream aspects of the assignment.

- **Assignee.** The assignee can be given access to AssignmentPro through the Mobile Employee Experience. In this streamlined portal, mobile employees can check the status of their assignments and manage related tasks and activities.
- **Service Providers.** It is rare for service providers (referred to as Vendors in AssignmentPro) to have direct integrations with AssignmentPro client instances. The most efficient way for service providers to interact with AssignmentPro data is through the Equus Ecosystem.

1.2 The Equus Hub

The Ecosystem is enabled by the **Equus Hub**, a central connection point for Equus clients and their connected service providers, as illustrated in Exhibit 1.

Exhibit 1: The Equus Hub manages secure communication



The primary job of the Equus Hub is to manage the secure communication of service-order-related messages between AssignmentPro instances and service-provider systems. For example, when a client managing an assignment sends a service order to a service provider, the service provider can receive the assignment details, fulfill services, and send updates back to the AssignmentPro client—without the user entering data manually.

1.2.1 API Method of Connection

For service providers connecting to the Equus Hub using the API Method, as described in this document, communication will occur automatically when:

- Service providers make web service requests to the RESTful endpoint hosted by the Equus Hub.
- The Equus Hub makes web service requests to the service providers' API endpoints.

The Equus Hub coordinates security and authentication for its inbound and outbound web service requests. Some aspects of the configuration of authentication, such as managing OAuth 2.0 credentials, require the service provider to access the Equus Hub's web interface manually.

 For more information, see the *Ecosystem API: Authentication Guide for Service Providers*.

1.2.2 Other Methods of Connection

The Equus Hub also enables two other methods of connecting service providers and AssignmentPro instances, as follows:

APro-to-APro Method



For service providers (such as relocation or assignment management companies) who use AssignmentPro to power their operations, the Equus Hub facilitates their connections to the AssignmentPro instances of their clients.

Connector Method



For service providers unable to integrate using web services and the API Method of connecting, they can access the Equus Hub's secure web interface and upload, send, and track service orders and associated updates for their AssignmentPro clients from a single location.

 Equus can provide information on the APro-to-APro Method and Connector Method of connecting to the Ecosystem, upon request.

1.2.3 Traffic Flow and Data Exchange

All traffic to and from the Equus Hub flows across secured HTTPS connections using TLS 1.2 encryption, with any of the supported connection methods. Also, service providers must be granted permissions to individual AssignmentPro instances before exchanging data with those instances through the Equus Hub.

Along with the capabilities described above, the Equus Hub can prevent service order requests from being routed when service provider systems are down for maintenance. To engage this functionality, a service provider must inform Equus of the timeframe the service provider API will be unavailable. When a service provider's system is identified as "**down with a maintenance window**," clients receive a message when they try to order services during this time. The service order communication is queued within AssignmentPro until the service provider's system is available again, at which time the stored communications are automatically sent.



Note: For security reasons, the Equus Hub, in normal operation, does not store the content of requests passing through it.

1.3 Service Provider Systems

Service providers are a crucial part of managing mobility programs. Service providers who connect their systems to the Ecosystem can:

- Automate communication, data entry, and workflows.
- Reduce duplicate information requests.
- Improve the accuracy of employee and service data.
- Communicate invoices to speed the payment cycle.
- Increase their clients' satisfaction.



1.3.1 Commonly Requested Service Types

The Ecosystem can support information exchange with compatible service-provider systems for the most commonly requested types of services that can be ordered from within AssignmentPro.

Some service providers may include multiple service types in a single initiated order or case. For these situations, multiple documents may need to be referenced to ensure the correct information is included in the data sets being communicated through the Ecosystem.

These supported service types and their associated supplementary documents are listed in the following table.

SERVICE TYPE	ASSOCIATED SUPPLEMENTARY GUIDE
Immigration and Visa Services	<i>Ecosystem API: Supplement for Immigration Service Providers</i>
Insurance Services	<i>Ecosystem API: Supplement for Insurance Service Providers</i>
Move Management Services • Household Goods Shipment • Vehicle Shipment • Pet Shipment • Insurance and Claims	<i>Ecosystem API: Supplement for Move Management Service Providers</i>
Relocation and Destination Services • Pre-Move Visit • Spousal/Family Assistance • Training • Area Orientation • School Search	<i>Ecosystem API: Supplement for Relocation and Destination Service Providers</i>
Tax Services • Tax Briefing • Returns • Equalization	<i>Ecosystem API: Supplement for Tax Service Providers</i>
Temporary Living Services • Needs Assessment • Housing Rental	<i>Ecosystem API: Supplement for Temporary Living Service Providers</i>

Equus can assist with determining how to combine the data sets and documentation that would be used. See section [4 Service Provider Onboarding and Connection Process](#) of this guide for more information.

To participate in the Ecosystem using the API Method of connection, service providers must expose web service endpoints to the Equus Hub. These endpoints provide connections either directly to the service providers' systems or through a gateway, middleware, or integration platform—providing access to the appropriate data within a service provider's internal systems.

1.3.2 Communication Endpoints

Endpoints are RESTful web service addresses in the service provider's system that allow it to receive service-related communication from AssignmentPro instances through the Equus Hub. The Equus Hub supports four general authentication methods for **connections made to a service providers'** RESTful web service endpoints, including OAuth 2.0, RSA, Mutual Authentication with X509 certificates, and Basic Authentication. The Ecosystem does not currently provide standard support for SOAP-based web service endpoints.

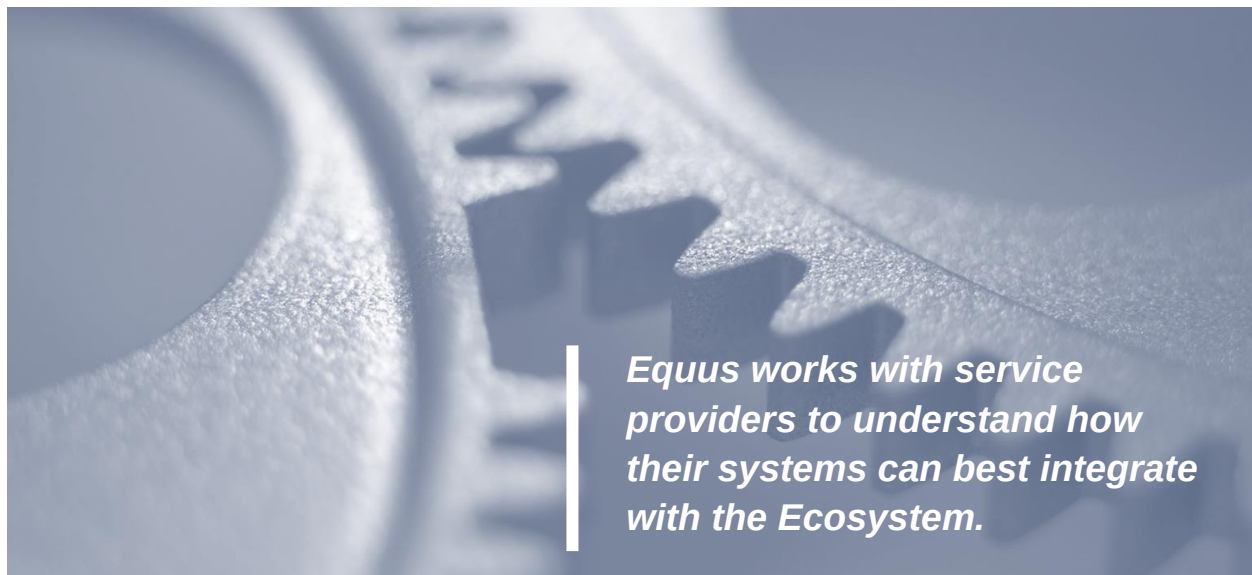
1.3.3 Connections to AssignmentPro

Service-provider systems must be able to connect to a RESTful web service endpoint and POST JSON-based payloads to be able to send information through the Equus Hub and then on to AssignmentPro instances. The Equus Hub supports three general authentication methods for **connections originating from service provider systems**, including OAuth 2.0, which Equus recommends as well as RSA and Mutual Authentication with X509 certificates.

 These four supported authentication options are discussed in more detail in the *Ecosystem API: Authentication Guide for Service Providers*.

Ecosystem integration provides the most value for all parties when two-way communication is enabled. However, service providers may only be able to connect one way (i.e., either receive initiations/follow-ups and cancellations only or push status and milestone updates only) or choose to implement bi-directional communication in a phased approach.

Equus works with service providers to understand how their systems can best integrate with the Ecosystem and assists with troubleshooting potential issues with integration connections and communication. See section [4 Service Provider Onboarding and Connection Process](#) of this guide for more information.



2 Ecosystem Integration Between Service Provider and AssignmentPro

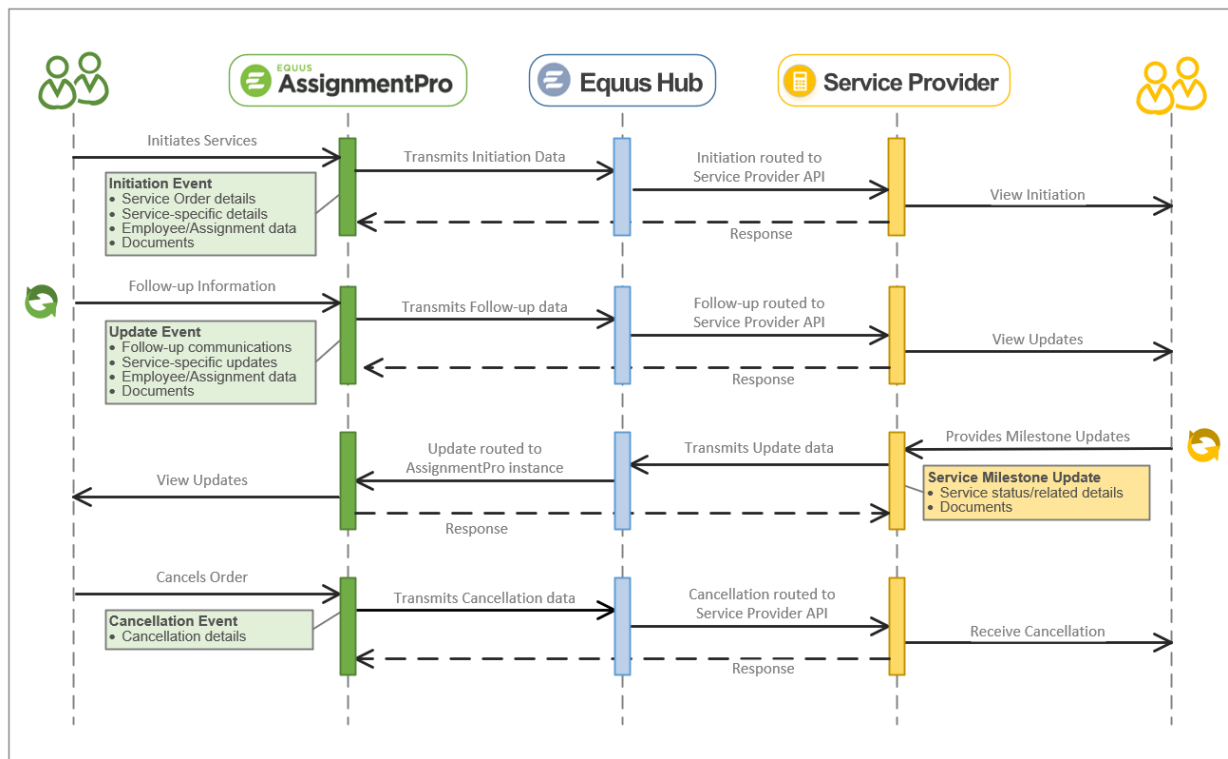
Ecosystem integrations between service providers and AssignmentPro client instances can involve two-way communication that flows in standard ways.

When a new request for provider services is initiated through AssignmentPro (i.e., a **Service Order**), the order is sent via web services to the service provider's system where a new case may be created. Ongoing updates to essential assignee and assignment information can also be sent to the service provider.

As the service provider works through the service delivery activities, real-time status and milestone updates can be sent back to AssignmentPro, allowing assignees and mobility team members to see the status of the service order.

The Ecosystem standard data flows, to and from the Equus Hub, are the means through which a service provider can seamlessly communicate with AssignmentPro. These data flows are illustrated in Exhibit 2 and described in more detail in the sections that follow.

Exhibit 2: Standard Ecosystem data flows



2.1 AssignmentPro to Service Provider

This section describes the standard data flows (i.e., processes and workflows), through the Equus Hub, from AssignmentPro to the service provider.

2.1.1 Standard Data Flows

The data flows in this direction consist of **Initiations**, **Follow-ups**, and **Cancellations**.

2.1.1.1 Initiations

When services need to be initiated for an assignee in AssignmentPro from a service provider, a **Service Order** is created, as shown in Exhibit 3, and related communication is sent. In AssignmentPro, the **Service Order** is the data object that represents the client's request for a service or services to be performed and provides the context for any communication related to the provision of those services. When the service order is sent, its status is set to "Ordered."

Exhibit 3: Example service order within AssignmentPro


Service Order Details
+ clientuser 5/16/2020
clientuser 5/16/2020

Status

Ordered

Service Order Number

7192

* Service Order Type

Relocation Services

* Vendor

ABC Services

Service Order Coordinator

Gloria Chu

Vendor Contact

Joe Veracruz

+

Service Due Date

Service Complete Date


Note/Communications
1

Show All

New Communication

Note Order Created: 5/16/2020 - By: clientuser

Please provide services for this VIP assignee


Expense Summary

New Voucher

Voucher	Policy	Account	Amount	Exch. Rate	Payment	Date	Reference
Save Close							

The initial communication sent from AssignmentPro to the Ecosystem-connected service provider will contain a collection of data fields in JSON format. The most important of these is the **Service Order ID**, which is the unique identifier given to the service order in AssignmentPro. The other data fields that are communicated with the initiation are determined during data mapping exercises.

Data mapping exercises are conducted by the Equus Integration Analyst, first with the service provider to establish an agreed data set and then with the client stakeholders, both as part of the initial onboarding and connecting to the Ecosystem. See section [4 Service Provider Onboarding and Connection Process](#) of this guide for more information.

Initiation Process Overview:

- Initiations are sent through the Equus Hub with **EventType <Initiation>** in the header.
- The service order's **Status** is set to "Ordered" after this event.

2.1.1.2 Follow-ups

Follow-ups are updates to the information sent to service providers after the service order has been initiated. Follow-up communication is triggered from the AssignmentPro instance to the service provider in one of two ways:

1. An AssignmentPro user sends a **Service Order Note/Communication** with the type "Follow Up." This communication sends the same set of data points that were sent to the service provider at the initiation of the service order.
2. Event-triggered updates are configured in the AssignmentPro instance, and the triggering event occurs. Event-triggered data updates refer to specific automation events that may be configured to push information to the service provider when specific data fields change in AssignmentPro. The data set that is communicated is typically the same as what was sent with the original order. However, the data set can also be configured to contain a different set of data fields. Examples of value-change triggers include:
 - a. The **Assignment Start Date** is updated.
 - b. A new **Cost Center** value is entered.
 - c. The assignee's **Marital Status** has changed.

Follow-ups Process Overview:

- Follow-ups are sent through the Equus Hub with **EventType <Update>** in the header.
- The service order's **Status** remains "Ordered."

If a service provider does not support follow-up updates, please coordinate with an Ecosystem Integration Analyst to determine if a "Not Implemented" or other error response to an **Update Event Type** is required.

2.1.1.3 Cancellations

A service order cancellation can be sent to the service provider if the service is no longer needed.

Cancellation Process Overview:

- Cancellations are sent through the Equus Hub with **EventType <Cancellation>** in the header.
- When the cancellation event has been sent successfully, the service order's status is set to "Cancelled."

If a service provider does not support cancellations, please coordinate with an Ecosystem Integration Analyst to determine if a "Not Implemented" or other error response to a **Cancellation Event Type** is required.

2.1.2 Request and Response Communication

A RESTful web service request will be made to the service provider's endpoint during the initiation of, follow up to, and cancellation of a service order. The initiation request will be made as an HTTP POST, and follow-up and cancellation requests can be made as either HTTP POST or PUT actions. The request must contain specific values in the HTTP header, and the request body or payload will be delivered in JSON format and with UTF-8 encoding.

2.1.2.1 Request Headers

In addition to the standard HTTP headers that would be included with a RESTful web service call, requests initiating from AssignmentPro will contain these additional values:

assignmentproinstanceid	This is the unique identifier of the AssignmentPro instance that initiated the web service request. Service-provider systems should validate this value to ensure the request is handled correctly for the specified client and environment.
assignmentprocompanyid	This is the identifier for the company configuration within an AssignmentPro instance. Corporate AssignmentPro clients typically have only one company designated in AssignmentPro; the value is usually "1." For relocation service providers and assignment management AssignmentPro clients, there can be many possible values to indicate their clients. This value should be validated when received.

vendorid

This is the unique identifier of the service provider's system that is being communicated with. **Service providers with multiple environments should validate that this value matches the expected value for the current environment.**

eventtype

This is the type of service order-related event that has triggered the request from AssignmentPro. As noted in the sections above, this value can be "Initiation," "Update," or "Cancellation."

Exhibit 4: Example request header

```
{
  "content-length": "3584",
  "assignmentproinstanceid": "a15a055a-b869-4108-a31e-138d80bf4d13",
  "assignmentprocompanyid": "1",
  "vendorid": "ec0575c6-97b1-11ea-bb37-0242ac130002",
  "eventtype": "Initiation",
  "accept": "application/json",
  "content-type": "application/json; charset=utf-8"
}
```

Additional headers will be included depending on the type of authentication (i.e., OAuth 2.0, RSA, Mutual Authentication with X509 certificates, or Basic Authentication), which is negotiated for Ecosystem connections to the service provider's endpoint.

 Authentication options are discussed further in the *Ecosystem API: Authentication Guide for Service Providers*.

2.1.2.2 Request Body

The data fields sent are organized and controlled through the Integration Data Mapping configuration in AssignmentPro. Integration Data Mappings, sometimes referred to as "push scripts," can support simple field-name masking. This configuration allows the service provider to receive uniform payloads from different AssignmentPro instances—containing data elements that may be stored in different locations within the clients' AssignmentPro instances. The configuration of the data mapping is performed during the service provider onboarding and connection process to a client AssignmentPro instances. See the [4 Service Provider Onboarding and Connection Process](#) section of this guide for more information.

Documents, images, and other files stored in AssignmentPro can be transmitted to the service provider as part of the request payload, like the example shown in Exhibit 5. Document files are transmitted as byte strings in hexadecimal format.

Exhibit 5: Example request payload

```
{
  "data": [
    {
      "name": "SERVICE_ORDER",
      "values": {
        "ID": 7192,
        "STATUS": "Ordered"
      }
    }, {
      "name": "NOTE_COMMUNICATION",
      "values": {
        "BODY_NOTE": "Please provide services for this VIP assignee"
      }
    }, {
      "name": "EMPLOYEE",
      "values": {
        "BIRTH_CITY": "Ann Arbor",
        "BIRTH_COUNTRY": "United States",
        "BIRTH_DATE": "1976-01-01T00:00:00",
        "FIRST_NAME": "Denise",
        "GENDER": "Female",
        "ID": 104365,
        "LAST_NAME": "Daniels",
        "MARITAL_STATUS": "Married",
        "PRIMARY_CITIZENSHIP": "United States",
      }
    }, {
      "name": "ASSIGNMENT",
      "values": {
        "COMPANY_ID": "Advanced Sync Logistics",
        "ID": 102325,
        "FROM_COUNTRY": "United Kingdom",
        "TO_COUNTRY": "United States",
        "TO_CITY": "New York",
        "START_DATE": "2021-01-01T00:00:00",
        "VIP": "Yes"
      }
    }, {
      "name": "DOCUMENT",
      "values": {
        "COPIED_FROM_DOCUMENT_ID": null,
        "CONTENT_LENGTH": 16256,
        "CONTENT_TYPE": "application/pdf",
        "DOCUMENT_TYPE_ID": "CV/Resume ",
        "DOCUMENT": "0x789C0BF06666116160646060A60C902D00C...",
        "ID": 11013,
        "NAME": "DanielsResume.pdf",
      }
    }
  ]
}
```

The Equus Hub can potentially apply additional payload transformation before routing requests from AssignmentPro instances to the service provider. This transformation capability can be used to facilitate connections to a service provider's system that would apply to connections in a general way; it is not meant to enable client-specific customizations. Service providers should confirm any additional transformation requirements with the Equus Integration Analyst, as some extensive transformations either may not be supported or may impact onboarding and connection project scope. Presently, the Equus Hub can make requests only to a single API endpoint to complete a single initiation, update, or cancellation. This capability will soon be expanded to support modern multi-endpoint APIs.

2.2 Service Provider to AssignmentPro

This section describes the standard data flow (i.e., process and workflows), through the Equus Hub, from the service provider to AssignmentPro.

2.2.1 Standard Data Flows

The standard data flows in this direction consist of status and milestone updates, and in limited circumstances, service provider invoice information.

2.2.1.1 Status and Milestone Updates

Service providers can communicate case-level information about the services they provide to the client and employee by pushing this information to the AssignmentPro instance that initiated the service order. These "Status and Milestone Updates" include updates about the:

- Delivery status of the service, including who to contact at the provider about the service
- Detailed information about the service being provided
- Service deliverables such as key dates or documents

The set of information provided in milestone updates will vary depending on the service type being performed, individual clients' AssignmentPro instance configurations, and the service provider's system capabilities. These parameters are determined during the data mapping collaboration as part of initial onboarding to the Ecosystem. See section [3 Ecosystem-Related Standard Data Sets](#) of this guide for more information.

These updates are allowed only when the AssignmentPro service order's "Status" is any of the following:

- Ordered
- Service complete
- Payment made
- Payment complete

The **Status** determines if the service provider can update the information associated with the service that was ordered.

2.2.1.2 Service Provider Invoice Support

If a client has configured AssignmentPro's invoice management features, then service providers may be able to send their invoices to AssignmentPro and submit them for review and approval. They may also be able to fetch invoice information from AssignmentPro. Equus helps service providers determine if invoices can be supported on specific Ecosystem connections.

2.2.2 Request and Response Communication

When communicating with an AssignmentPro instance, the service-provider system must make a web service request to the Equus Hub's RESTful web service endpoint. This request must be made as an HTTP POST action, and the request body or payload should be delivered in the Equus-specified JSON format called the Equus Protocol for Inter-process Communication or EPIC. EPIC is the SQL-like scripting language for interacting with the integration API exposed by AssignmentPro instances and accessed through the Equus Hub. Equus Integration Analysts will work with service providers to ensure that the proper EPIC scripts are developed for specific client and service provider use cases.

2.2.2.1 Request Headers

In addition to the standard HTTP headers typically included with a RESTful web service call, requests initiating from a service provider's system to an AssignmentPro instance should contain these additional values:

content-type	The POST body of requests sent to AssignmentPro should consist of UTF-8-encoded JSON data with a value of either "application/json" or "application/json; charset=utf-8."
vendorid	This required header's value is the unique identifier of the service provider's system.
assignmentproinstanceid	This header's value is the unique identifier of the AssignmentPro instance that initiated the web service request. This header and value are required.

Exhibit 6: Example required incoming headers

```
"content-type": "application/json"
"vendorid": "2c681dc7-09fe-4068-8e42-6f3c21ffb4cd",
"assignmentproinstanceid": "09b18e2b-7f06-4dd9-9049-2ac1d75761d1",
```

2.2.2.2 Request Body

The body of requests made by service providers should contain a valid JSON-based EPIC script with UTF-8 encoding.

Exhibit 7 is an example of a basic EPIC script to set the **Vendor Status** field on a service order to "Accepted." The **Vendor Status** is a 100-character text field, which is set to a status the service provider considers important to report. Note the similarities with SQL, such as the command "UPDATE" or the "whereClause" element.

Exhibit 7: Example JSON script to set the vendor status field to "Accepted"

```
{
  "command": "UnitOfWork",
  "singleTransaction": true,
  "commandList": [
    {
      "command": "UPDATE",
      "targetTable": "SERVICE_ORDER",
      "targetVariable": "SERVICE_ORDER!ID",
      "targetProperty": "SERVICE_ORDER!ID",
      "whereClause": "[ID]=@SERVICE_ORDER_ID",
      "whereParams": [
        {
          "name": "@SERVICE_ORDER_ID",
          "value": "11319"
        }
      ],
      "values": [
        {
          "name": "VENDOR_STATUS",
          "value": "Accepted"
        }
      ]
    }
  ]
}
```

The service provider is required to provide the **Service Order ID** as an EPIC variable when sending updates. AssignmentPro uses this value and the applicable permission set (described in section [3.4 Permission Sets](#)) to establish the service provider's permissions to access and update employee, assignment, and service-related data.

2.2.2.3 JSON-to-EPIC Transformation

The Equus Hub can apply JSON-to-EPIC payload transformation before requests from service provider systems are routed to an AssignmentPro instance. Please confirm any additional

transformation requirements with your Equus Integration Analyst as some extensive transformations either may not be supported or may impact onboarding and connection project scope and timelines.

2.2.2.4 Other Requests

Service providers should confirm any requirements outside of the presented standard flows with the Equus Integration Analyst. The standard data flow patterns presented in this document are designed to minimize the amount of custom development and configuration effort required from service providers and Equus to make multiple connections between Ecosystem service providers' systems and multiple clients' AssignmentPro instances. For example, alternative data flows, where the service-provider system would poll instances periodically for new service orders, are generally not supported.

2.2.2.5 Successful Responses

Successfully processed requests will receive an HTTP 200 response with information about the related AssignmentPro records, like the example shown in Exhibit 8.

Exhibit 8: Example of a successfully processed request

```
{ "Result": { "SERVICE_ORDER!ID": 7192 },  
  "ErrorMessages": [],  
  "IsValid": true }
```

2.2.3 Error Responses and Troubleshooting

There are a variety of error responses that can be returned from the Equus Hub when a service provider makes a web service request. These error response types include:

- Connection
- Access and Permission
 - Access denied or non-existing records
 - CRUD
- Message Parsing Errors
 - EPIC Syntax/JSON Mapping
 - JSON Syntax
- Validation and Processing
 - Required Field Validation
 - Invalid Lookup Value

Descriptions and examples, if applicable, of each error response type are provided in the sections that follow.

 **ERRORS RETURNED FROM** the Equus Hub, AssignmentPro instances, and service provider systems may contain Personally Identifiable Information (PII). Equus and service provider partners must take all appropriate precautions to protect the data of our shared clients. Errors should not be logged in plain text or sent via email without filtering, obfuscation, or encryption.

2.2.3.1.1 Connection Errors

If the call to the Equus Hub or AssignmentPro fails, a standard HTTP connection response will be returned to the caller. Examples of standard error codes and associated response definitions are listed in Exhibit 9.

Exhibit 9: Standard HTTP connection error codes

HTTP RESPONSE CODE	POSSIBLE RESPONSE DEFINITION
400	Bad Request – Syntax error in HTTPS call
401	Unauthorized – Caller does not have valid authentication information
403	Forbidden – Caller does not have access to make requests
404	Not Found – Unable to find the endpoint
500	Server Error – Contact technical support

2.2.3.1.2 Access and Permissions Errors

If a service provider does not have access permissions to a particular set of data, associated table, or method, an "Access Denied" error will be returned to the caller when the invalid request is made.

Access Denied or Non-Existing Record Errors. When a command is sent to AssignmentPro for an ID value that is either not accessible by the service provider or non-existent in the system, an error message will be returned to the caller like the example shown in Exhibit 10.

Exhibit 10: Example "Access Denied" or "Non-Existing Record" error message

```
{
  "Result": null,
  "ErrorMessages": [
    "There was an error trying to retrieve service order information from AssignmentPro:
      The status was: ProtocolError.
      The status description was: Internal Server Error.
      The response body was: {"
```

```

        \"ERROR\": \"Errors returned from server\",
        \"TechnicalErrors\": [{
            \"ErrorContextInformation\": \"\",
            \"ExceptionMessage\": \"The parent record referenced by
'DOCUMENT_ASSIGNMENT_VISA!ATTACHED_RECORD_ID' with value: '54321' does
not exist or you do not have rights.\",
            \"ExceptionType\": \"\",
            \"Message\": \"Unexpected error: The parent record
referenced by 'DOCUMENT_ASSIGNMENT_VISA!ATTACHED_RECORD_ID' with value:
'54321' does not exist or you do not have rights.\"}],
        \"ValidationErrors\": []
    .],
    \"IsValid\": false}

```

CRUD Errors. When a service provider does not have access to create, read, update, or delete (CRUD) a record, an error message will be returned to the caller, like the example shown in Exhibit 11.

Exhibit 11: Example CRUD error message

```

{\"Result\": null,
 \"ErrorMessages\": [\"There was an error trying to retrieve servicer order
information from AssignmentPro:
    The status was: ProtocolError.
    The status description was: Internal Server Error.
    The response body was: {
        \"ERROR\": \"Errors returned from Server\",
        \"TechnicalErrors\": [{
            \"ErrorContextInformation\": \"\",
            \"ExceptionMessage\": \" Denied CRUD right 'Update' for
access to 'VENDOR_CONTACT' with field list: 'VENDOR_CONTACT!MOBILE_PHONE,
VENDOR_CONTACT!JOB_TITLE,' and where clause: '[EMAIL] = @EMAIL AND
[VENDOR_ID] = @VENDOR_ID' and parameters: '
@EMAIL:\\\\\"todd.smith@mailinator.com\\\\\", @VENDOR_ID:\\\\\"13952\\\\\", ',
and orderby clause: '' and orderby parameters: ''. \",
            \"ExceptionType\": \"\",
            \"ExceptionMessage\": \"Update: Unexpected error: Denied
CRUD right 'Update' for access to 'VENDOR_CONTACT' with field list:
'VENDOR_CONTACT!MOBILE_PHONE, VENDOR_CONTACT!JOB_TITLE,' and where
clause: '[EMAIL] = @EMAIL AND [VENDOR_ID] = @VENDOR_ID' and parameters: '
@EMAIL:\\\\\"todd.smith@mailinator.com\\\\\", @VENDOR_ID:\\\\\"13952\\\\\", ',
and orderby clause: '' and orderby parameters: ''. \"}],
        \"ValidationErrors\": []
    .],
    \"IsValid\": false}

```

This type of error could be caused by multiple reasons, such as:

- The **Vendor** does not have CRUD rights to a record. An invalid "whereClause" was used in the JSON command.
- The **Vendor** is no longer active.
- The **orderby** clause in the JSON command is invalid.

2.2.3.1.3 Message Parsing Errors

If a request is made that contains EPIC or JSON formatting or mapping problems, an error message will be returned based on the syntax issue.

EPIC Syntax/JSON Mapping Errors. A JSON-mapping error will be sent if a request with an invalid parameter is made within the EPIC request. When this occurs, an error message will be sent to the service provider's endpoint, like the example shown in Exhibit 12.

Exhibit 12: Example EPIC Syntax/JSON Mapping error message

```
{ "Result":null,
  "ErrorMessages" :[ "There was an error trying to retrieve service order
information from AssignmentPro:
    The status was: ProtocolError.
    The status description was: Internal Server Error.
    The Response body was: {
      \"ERROR\": \"Errors returned from server\",
      \"TechnicalErrors\": [{
        \"ErrorContextInformation\": \"JSON Map Error.\",
        \"ExceptionMessage\": \"JSON Property 'wherClause' is
not defined in the class
'Equis.Grow.SupportLayer.EPIC.Instructions.GET'.\",
        \"ExceptionType\": \"\",
        \"Message\": \"JSON Property 'wherClause' is not defined
in class 'Equis.Gro.SupportLayer.EPIC.Instruction.GET'.\"}],
      \"ValidationErrors\": []}
    .\"],
  \"IsValid\":false}
```

This error message can occur when:

- A parameter is misspelled (e.g., in the example above, the "whereClause" parameter is misspelled as wherClause).
- A parameter is used that does not belong with the table being referenced.
- An invalid command is used.

JSON Syntax Errors. If the structure of a command is written in the incorrect JSON format, an error message will be sent to the service provider's endpoint. Information within the error message will indicate the location of syntax error within the JSON command and what characters should correct the JSON. When this error occurs, an error message will be returned, like the example shown in Exhibit 13.

Exhibit 13: Example JSON Syntax error message

```
{ "Result":null,
  "ErrorMessages" :[ "There was an error trying to retrieve service order
information from AssignmentPro:
    The status was: ProtocolError.
    The description was: Internal Server Error.
    The response body was: {
      \"ERROR\": \"Errors returned from server\",
      \"TechnicalErrors\": [{
        \"ErrorContextInformation\": \"JSON Parse Error.\",
        \"ExceptionMessage\": \"JSON Syntax error at position
226 (position 16 in '\\\\\"name\\\\\\\":\\\\\\\"STATUS\\\\\\\":). Encountered ':'.
Expected ', ' to delimit an element of the array, or ']' to terminate the
array.\",
        \"ExceptionType\": \"\",
        \"Message\": \"JSON Syntax error at position 226
(position 16 in '\\\\\"name\\\\\\\":\\\\\\\"STATUS\\\\\\\":). Encountered ':'. Expected
', ' to delimit an element of the array, or ']' to terminate the
array.\"}],
      \"ValidationErrors\": []}
    .\"],
  \"IsValid\":false}
```

2.2.3.1.4 Validation and Processing Errors

Validation and processing errors occur when invalid values are given, or missing required values are included in the JSON command.

Required Field Validation Errors. When a required value is missing from a JSON command script, an error message will be returned, like the example shown in Exhibit 14.

Exhibit 14: Example error message for a missing required field in a JSON command script

```
{ "Result":null,
  "ErrorMessages" :[ "There was an error trying to retrieve service order
information from AssignmentPro:
    The status was: ProtocolError.
    The status description was: Internal Server Error.
    The response body was: {
```

```

        \ "ERROR\":\ "Errors returned from server\ ",
        \ "TechnicalErrors\":[{
            \ "ErrorContextInformation\":\ "\ ",
            \ "ExceptionMessage\":\ "Both DOCUMENT!DOCUMENT and
DOCUMENT!FILE_NAME are required when either are given.\ ",
            \ "ExceptionType\":\ "\ ",
            \ "Message\":\ "Unexpected error: Both DOCUMENT!DOCUMENT
and DOCUMENT!FILE_NAME are required when either are given.\ "}],
        \ "ValidationErrors\":[]}]
    ."],
    "IsValid":false}

```

Invalid Lookup Value Errors. When a value is a lookup value, and the exact lookup code is not used within the script, the **"valuelookup"** parameter must be included. When this value is included, it searches for values within the database that match the supplied value.

If the value does not match any values within the database, an error message will be returned, like the example shown in Exhibit 15.

Exhibit 15: Example error message for an invalid lookup value

```

{"Result":null,
"ErrorMessages" :[ "There was an error trying to retrieve service order
information from AssignmentPro:
    The status was: ProtocolError.
    The status description was: Internal Server Error.
    The response body was: {
        \ "ERROR\":\ "Errors returned from server\ ",
        \ "TechnicalErrors\":[{
            \ "ErrorContextInformation\":\ "\ ",
            \ "ExceptionMessage\":\ "No data value found for lookup
value 'Misc' in field: DOCUMENT!DOCUMENT_TYPE_ID\ ",
            \ "ExceptionType\":\ "\ ",
            \ "Message\":\ "Unexpected error: No data value found for
lookup value 'Misc' in field: DOCUMENT!DOCUMENT_TYPE_ID\ "}],
        \ "ValidationErrors\":[]}]
    ."],
    "IsValid":false}

```

3 Ecosystem-Related Data Sets



In the Equus Ecosystem, a "data set" refers to the collection of data fields and document types that could be transmitted as part of Ecosystem integrations between AssignmentPro and a service provider's system.

Ecosystem data sets include the fields of data and document types that are typically pushed from AssignmentPro as part of the [Service Order Initiations/Follow-Ups and Cancellations](#) data flows. These data sets also include the fields, milestone statuses, and document types that can be sent from the service provider to AssignmentPro.

Within the Ecosystem, there are three types of data sets to facilitate the transfer of data from AssignmentPro to the service provider's system:

3.1 Recommended Service

The "Recommended Service" data set contains the fields and document types that Equus considers the best-in-class set for a service type that is supported by AssignmentPro's standard configuration. This data set represents the optimal data collection practices from leading service providers in the industry.

Equus recommends this set of data for exchange between AssignmentPro and the service provider's system because it will provide the best value to a client if implemented by the service provider.

3.2 Service Provider-Agreed

The "Service Provider-Agreed" data set presents the fields, document types, and milestone statuses that a service provider has agreed to exchange with any client using AssignmentPro. This data set will include data elements from one or more "Recommended Service" data sets that the service provider's system can support.

The goal of the Service Provider-Agreed data sets is to reduce the effort involved in connecting an Ecosystem service provider to multiple AssignmentPro client instances with as little per-client field mapping and configuration effort as possible.

3.3 Client/Service Provider-Agreed

The "Client/Service Provider-Agreed" data set represents the fields, document types, and milestone statuses that a service provider has agreed to exchange with a specific

AssignmentPro client. This data set will include data elements from the "Service Provider-Agreed" data set. It may contain information about field name masking or lookup value transformations that are required to accommodate specific client implementations of AssignmentPro and configuration in the service provider's system.



WIDE DISCREPANCIES BETWEEN the Client/Service Provider-Agreed data set and the Service Provider-Agreed data set increases the duration and complexity around creating and maintaining the connection.

For this reason, including additional standard fields, custom fields, or user table fields is not typically supported. Equus may have the ability to conform data sets to service providers' formats; however, even when possible, this process increases the scope, complexity, and investment of the project implementation.

3.4 Permission Sets

Any EPIC request made to an AssignmentPro system is subject to multiple layers of data security, the primary of which is permission sets. Permission sets consist of allowed or disallowed actions, fields, tables, 'where' clauses, and methods.

Equus has standard permission sets that cover each of the Ecosystem-enabled service types, and in general, service providers do not need to be concerned with permission set details. The standard permission sets will be used with any connection using either a "Recommended Service" data set in abstract or a "Service Provider-Agreed" data set in practice.

The circumstances where these permission sets cannot be used are when:

- A Client/Service Provider-Agreed data set contains fields or actions that are outside the bounds of the standard permission set.
- There are requirements to limit the boundaries more tightly of what a service provider can do with EPIC scripts.

In these cases, a custom permission set may need to be created by Equus and implemented in AssignmentPro to support the service provider connections with one or more client AssignmentPro instances. However, this is not typical and may affect the onboarding and connection project scope.

4 Service Provider Onboarding and Connection Process

An Equus Integration Analyst is assigned to facilitate and support the standard process for onboarding a new service provider to the Ecosystem network and connecting that service provider to their first or subsequent client AssignmentPro instance.



THE INFORMATION IN THIS SECTION COVERS THE MAJOR STEPS AND ACTIVITIES INVOLVED IN THE ONBOARDING AND CONNECTION PROCESS, FROM A HIGH-LEVEL PERSPECTIVE.

Creating robust connections between service-provider systems and the Ecosystem is critical for clients and service providers to receive the value of being part of the Equus Ecosystem. Equus collaborates closely with service providers to:

- Communicate the business value of integrating with AssignmentPro.
- Understand the needs in general of the service provider's AssignmentPro clients.
- Ensure an understanding of the development work required to implement the necessary integrations successfully.

4.1 New Ecosystem Service Provider Onboarding

The Equus Integration Analyst will provide documentation to the service provider as well as schedule a kickoff meeting with Equus, the service provider's technical personnel, and business stakeholders. The goal of the onboarding process is for Equus and the service provider to create Ecosystem integrations that can benefit multiple current and future clients. For this reason, client representatives are generally not involved in these steps.

4.1.1 *New Service Provider Kickoff Meeting*

The purpose of this kickoff meeting is to familiarize the service provider with the Ecosystem's setup process and technical considerations. During this meeting, the participants will:

- Review Ecosystem documentation to ensure documentation is prescriptive.
- Review and determine authentication methods for each integration direction.
- Review the Recommended data set for the type of service of services and answer questions.
- Determine any reference data load or transformation requirements.
- Discuss concerns, problems, and gaps between the Equus standard and what the service provider can support.
- If not covered separately, discuss shared clients, their needs, and integration timelines.

At the meeting's conclusion, further meetings and communications will be planned to cover data mapping collaboration.

4.1.2 Data Mapping Collaboration

The data mapping collaboration will be a series of meetings between the Equus Integration Analyst and the service provider's team to refine integration requirements. It is important at this stage to identify what is supportable for shared clients, generically, and flag requirements that are not within the supported interface capabilities.

After Equus and the service provider complete the data mapping collaboration, a Service Provider-Agreed data set will be produced. This data set will be sent to each client team that will be connected to the service provider through the Ecosystem in the current project scope. The Service Provider-Agreed data set will then be reviewed and adapted to the clients' specific AssignmentPro configurations and service provider variation for those clients.

4.1.3 Development

Based on the requirements that have been discussed and agreed on, the service provider and Equus will develop the components required to complete the integration.

4.1.4 Initial Service Provider Connection and Testing

The initial connection here is referring to the process of initiating a request so the Equus Hub can successfully contact, authenticate, and send requests to the service provider's API. It also means ensuring the service provider can authenticate in the Equus Hub and send status and milestone updates requests to AssignmentPro. The primary activities at this stage include:

- | | |
|---|--|
|  | The Equus Integration Analyst will configure the service provider in the Equus Hub and communicate the service provider's vendorid . |
|  | The Equus Integration Analyst will configure the service provider as a Vendor within an Ecosystem sandbox instance of AssignmentPro and communicate the instance's assignmentproinstanceid and assignmentprocompanyid . |
|  | Equus and the service provider will thoroughly test communication to and from the service provider's system. |
|  | Equus and the service provider will review and troubleshoot any errors and issues that are found during testing. |

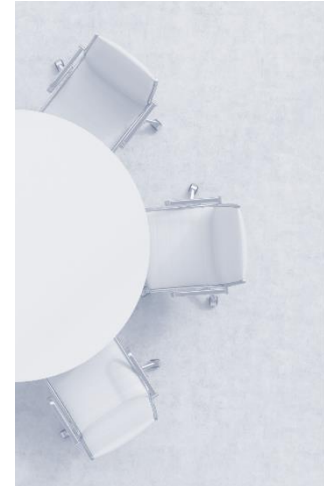
The outcome of this step should be thoroughly tested connections and communication between AssignmentPro and the service provider system, based on an agreed and fully tested Service Provider-Agreed data set.

4.2 Service Provider to Client AssignmentPro Instance Connection and Testing

These steps cover the service provider-to-client connection process and would be repeated for each client AssignmentPro instance being connected. For service providers who are new to the Ecosystem, scheduling the preliminary meetings and data mapping collaboration described below may overlap with some of the onboarding steps.

4.2.1 Pre-Kickoff Meeting

The Equus Integration Analyst will schedule a pre-kickoff meeting with the Equus Ecosystem team, the Equus client team, the service provider's main project contact(s), the service provider's client engagement team, and any other service-provider stakeholders as deemed appropriate.



In this meeting, attendees will share information about the client's needs and work through any potential issues that might come up when meeting with the client in the Client/Service Provider Kickoff meeting.

4.2.2 Client/Service Provider Kickoff Meeting

Following the pre-kickoff meeting, the Equus Integration Analyst will schedule the actual kickoff meeting with the appropriate client, service provider, and Equus representatives. The goal of this meeting is for all parties to understand the roles they will play on the project and set expectations regarding the project steps, communications, and timeline.

4.2.3 Data Mapping Collaboration

Starting with the existing Service Provider-Agreed data set, the Equus, client, and service provider teams will build a Client/Service Provider-Agreed data set that aligns with the client's specific service ordering workflow and AssignmentPro configuration.

4.2.4 Initial Client Instance Connection and Testing

This process is similar to the configuration and connection steps that were described above in section [4.1.4 Initial Service Provider Connection and Testing](#). The Equus Integration Analyst will provide the `assignmentproinstanceid` and `assignmentprocompanyid` for the client AssignmentPro instance that is being connected. The service provider will use these values with their existing `vendorid` to configure their system to support communication with the client's AssignmentPro instance.

The Equus client team will work with the client to complete acceptance test cases that will confirm the quality of the connection and communications. Errors at this stage should be rare since testing had already been completed between the service provider system and an Ecosystem sandbox AssignmentPro instance.

SUMMARY

This guide has provided information regarding the ecosystem platform and its components, the integration and standard data flows between web services and AssignmentPro, an introduction to the Ecosystem-related data sets, and an overview of the process for onboarding and connection service providers who are new to the Ecosystem.

Rich data, combined with innovative technology tools, helps connect the entire mobility value chain including service provider systems to facilitate the seamless management of a company's global workforce. Through the Equus Ecosystem, the entire business has visibility into the move status, compliance, costs, and program demographics—all consolidated with performance, KPIs, and employee skills and experiences.

Harness the power of connectivity to simplify the mobility process, gain strategic insights, and enhance the employee experience with the Equus Ecosystem. Global Mobility. Connected.

ABOUT EQUUS

Equus Software is the global leader in cloud-based international relocation and mobility solutions. More than 300 organizations around the world rely on Equus tools and technology to automate mundane, transactional work so that global mobility teams, talent management professionals and other key stakeholders can focus on adding value to the business. Founded in 1999, Equus has a proven track record for delivering cutting-edge talent mobility solutions, continuous innovation and exceptional customer service.



Inquiries@equusoft.com
+1 (303) 292-4200
www.equusoft.com